DialReader: Project Plan

Oct 12, 2026 · @Tejas

Summary

DialReader reads an analogue dial gauge from one phone photo using classical image processing, and uses GenAI at only three narrow points where the classical pipeline is weak or manual.

The IPCV pipeline does all the measuring. It restores the photo, finds the dial face, undoes the camera tilt, unwraps the dial into a straight strip, finds the tick marks and the needle, and converts the needle angle into a value with an error bar. When the result is unreliable, it says so instead of guessing.

The rule for GenAI: a hook is allowed only if it plugs into a weak or manual step, its output is checked by classical measurements, and a before/after ablation shows whether it helps. The pipeline must still run with every hook switched off.

- **Scale parser:** a vision-language model (VLM) reads the dial face (range, unit, scale type, printed labels), and geometry validates that parse against the detected ticks.
- **Gross-error sentinel:** a VLM gives a coarse reading, and a large disagreement with the geometric reading raises a flag.
- **Repair loop:** on low confidence, an LLM looks at diagnostic overlays and picks one fix from a fixed menu. It never produces a number.

What you will deliver:

- A working pipeline and a Streamlit demo: upload a photo, see every stage, get the reading with its error bar and a reliability flag.
- A dataset of about 300 of your own photos with ground truth, plus a test on the public MeasureBench dial subset.
- Experiments: stage ablations, robustness sweeps, a five-step GenAI ablation ladder and abstention curves.
- Coverage of all five syllabus units and all 14 lab weeks, with each weekly program serving as one module of the project.

The problem and what already exists

Frontier VLMs read real instruments poorly, and earlier dedicated pipelines do not transfer to unseen gauge types, which leaves room for a careful classical-first design with measured GenAI help.

Pressure gauges, water meters, thermometers and kitchen scales have no digital output. A phone photo that turns them into readings is useful in factories, labs and farm pumps, and for retrofitting old equipment without sensors.

Evidence that the problem is still open

- MeasureBench has 2,442 image-question pairs, and its real subset covers 26 instrument types. The best of 18 VLMs scored 30.2% overall on real images. Unit recognition was above 90%, but value accuracy was only 30.7%, because models misplace the pointer. The models tested were from 2025, so newer ones may do better.
- A 2026 study of gauge videos with GPT-5.x and Gemini 3 models found limited ability to interpret needle trajectories and scale semantics (based on its abstract).
- Two earlier special-purpose systems (Reitsma et al. 2024 and Shu et al. 2023) generalised poorly on MeasureBench: 8.5% overall accuracy for the first, and 4.2% value accuracy on real images for the second.
- In a vehicle-dashboard benchmark, ellipse fitting stayed more robust than learned models when gauge masks were resized between 1/4 and 4 times. This supports a geometry-first core.

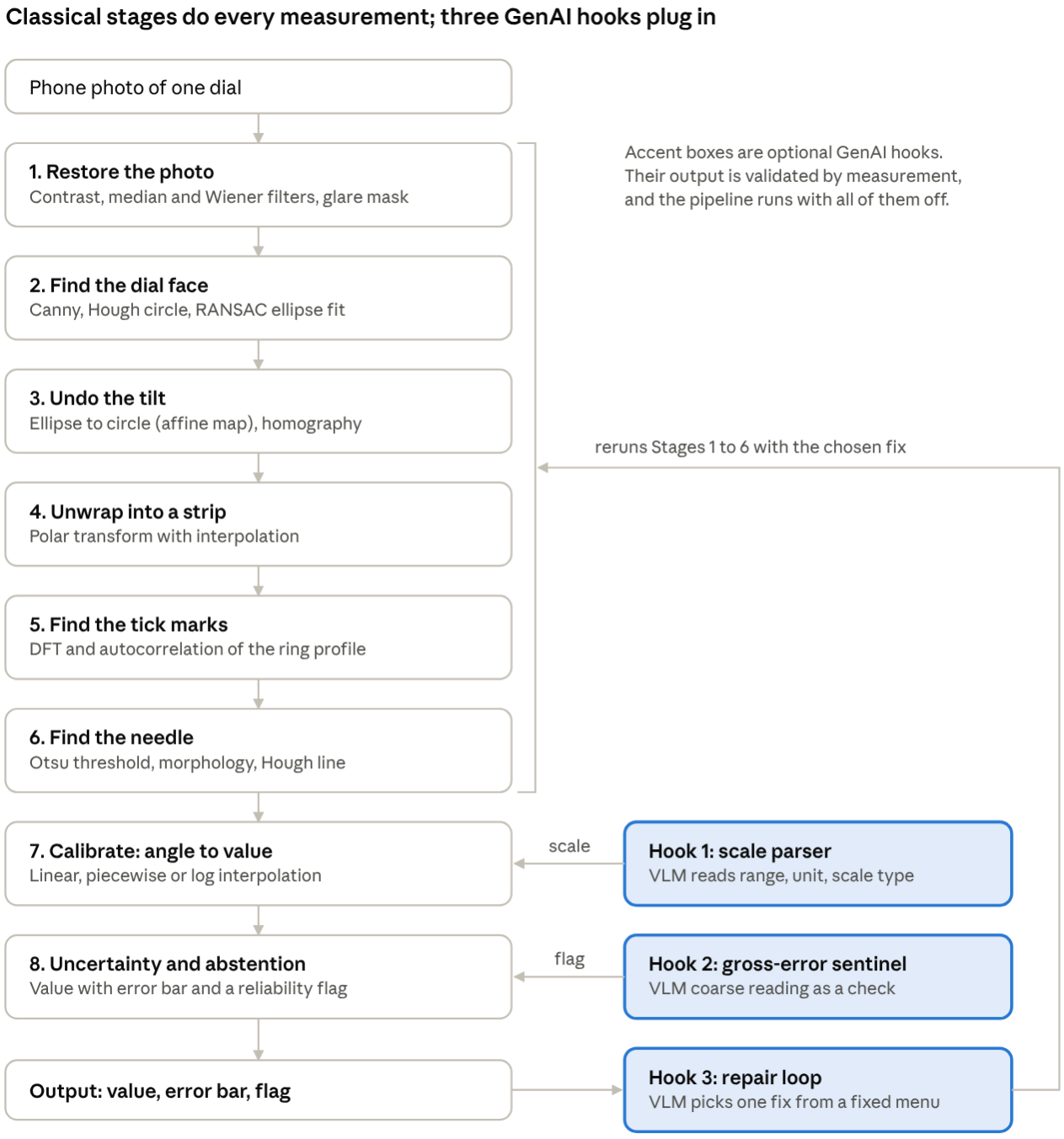
Related work

- Leon-Alcazar et al., WACV 2024: a two-stage CNN trained on Blender-rendered synthetic gauges, with a real test set of 4,813 images.
- Reitsma et al., ICRA 2024: gauge detection, notch detection, ellipse fitting, needle segmentation and OCR of the scale. It assumes a linear scale with at least five visible major notches.
- DialBench: a pointer-meter benchmark on which 18 multimodal models were retrained, plus a VLM that injects physical relations.
- Federated meter reading: unwraps dials into a linear tick ruler with a polar transform.
- River gauge plate reading: combines object detection with GPT-4o and Gemini.

Where this project sits. No published system was found that does all three of these: it is classical-first with a DFT-based tick check and a reading with an error bar; it uses a VLM only to parse the scale and checks the parse against detected ticks; and it tests each GenAI step by ablation against a classical-only pipeline. The search was not exhaustive, so word the claim in your report as "no prior work found", not "nobody has done this". Each classical technique used here (Hough, ellipse fitting, polar unwrapping) is established, so the contribution is the combination and the measurement.

System overview

The pipeline has eight core stages built from IPCV techniques, with three GenAI hooks attached at its weak points.



pipeline flow · 8 stages, 3 GenAI hooks

Read it top to bottom: the photo flows through Stages 1 to 8 on the left. Hook 1 feeds Stage 7, Hook 2 feeds the reliability flag of Stage 8, and Hook 3 reruns Stages 1 to 6 when the output is unreliable. Stages 9 to 11 (gauge type, registration, compression) are optional extensions outside this flow.

Stages 1 to 4: from a messy photo to a straight strip

These four stages turn a phone photo into a rectified, unwrapped strip in which tick marks and the needle become simple one-dimensional features.

Stage 1: Restore the photo

Real photos have uneven light, motion blur, noise and glare on the glass cover.

- **Contrast:** histogram equalisation, and its adaptive form CLAHE (an extension of histogram processing) for dim or unevenly lit dials.
- **Noise:** Gaussian smoothing for Gaussian noise and a median filter for salt-and-pepper noise (a linear and a non-linear filter).
- **Sharpening:** unsharp masking or the Laplacian.
- **Deblurring:** a Wiener filter, which is a least-mean-square restoration filter. The blur kernel is unknown, so try a small set of Gaussian and linear-motion kernels and keep the one that gives the highest sharpness score (variance of the Laplacian). The DFT of a blurred image also shows the kernel's pattern of zeros, which is a useful cross-check.
- **Glare:** mask saturated pixels by thresholding brightness, then close and dilate the mask with morphology. Later stages treat masked pixels as missing, and heavy glare on the needle lowers confidence.
- **Colour:** keep a colour copy, because many needles are red and one colour channel can separate the needle better than grey levels.

To test it, add known blur and noise to clean photos, then compare PSNR and the final reading error with and without each step.

Stage 2: Find the dial face

- Canny edge detection (Gaussian smoothing, gradient, non-maximum suppression, hysteresis thresholding) gives the edge pixels.
- The Hough circle transform votes in (x, y, r) space and gives a coarse circle even when part of the rim is missing.
- A tilted dial appears as an ellipse, so take the edge points near that circle and fit an ellipse (centre, two axes and an angle) with RANSAC. RANSAC ignores outlier edges from text, shadows and the bezel.
- The output is the ellipse and its inlier ratio, which becomes the first confidence signal.
- Failure cases are low-contrast rims and bezels that create several concentric ellipses. Choose the ellipse that encloses the tick marks.

Stage 3: Undo the tilt

A circle seen at an angle becomes an ellipse, so mapping the fitted ellipse back to a circle approximately restores the frontal view. Shift the centre to the origin, rotate the major axis onto the x-axis, scale the minor axis to match the major axis, and rotate back. This is an affine transform. It is accurate for moderate tilt and only approximate under strong perspective. If the gauge has a square or rectangular bezel, its four corners give a full homography instead.

To test it, measure reading error at tilts of 0, 20 and 40 degrees, with and without this stage.

Stage 4: Unwrap the dial into a strip

The polar transform samples the rectified image around the dial centre with bilinear interpolation. The horizontal axis of the strip is the angle (for example 0.25 degrees per column, giving 1,440 columns) and the vertical axis is the radius.

```
S(\theta, \rho) = I\left(c_x + \rho\cos\theta,\; c_y + \rho\sin\theta\right)
```

In the strip, tick marks become short vertical marks at regular spacing and the needle becomes one vertical line. Two 2D problems turn into one-dimensional peak finding.

The centre must be accurate, because a centre error bends straight features into wavy curves. Refine it by searching a few pixels around the fitted centre for the position that makes the tick profile of Stage 5 sharpest.

Stages 5 to 8: from the strip to a reading with an error bar

These stages find the ticks and the needle in the strip, convert the needle angle into a value, and decide whether the result can be trusted.

Stage 5: Tick marks and the scale geometry

- Collapse the radial band where the ticks lie into a one-dimensional profile P(theta), the mean darkness over radius. The band is the range of radii with the strongest periodic contrast.
- Ticks are periodic in angle. Peak detection gives tick positions, and the DFT or autocorrelation of P(theta) gives the tick spacing. Ticks usually cover only an arc (for example 270 degrees), so restrict to that arc and apply a window to limit spectral leakage, or use autocorrelation, which copes better with partial coverage.
- Fit a regular comb (spacing and offset) to the detected peaks. A large residual means a missed or false tick and lowers confidence.

- Major and minor ticks differ in length, meaning their extent along the radius.
- Optional checks: a Haar wavelet decomposition of P(theta) to denoise it, and Harris corners or LoG blobs at the tick ends as an independent detector.
- Output: tick angles labelled major or minor, the start and end angles of the scale, and the tick count.

Stage 6: Needle angle

- Segment the needle with an Otsu threshold on the strip, or on a colour channel when the needle is red. Opening removes specks left by ticks and text, and closing joins broken needle pieces.
- In the strip the needle is a vertical band. Summing the mask over the radii inside the tick ring gives a column profile whose peak is the needle angle. Refine it with a line fit to the thinned needle (a morphological skeleton).
- Cross-check with a Hough line through the centre in the rectified image. The two estimates must agree within a tolerance, otherwise confidence drops.
- Tell the tip from the tail by comparing how far the needle extends on each side of the centre. The longer side is the tip.
- Output: the needle angle and its spread, taken from the width of the profile peak or the line-fit residual.

Stage 7: Calibration (angle to value)

Calibration needs the start and end angles of the scale, the values at both ends and the direction of increase. For a linear scale, with angles measured along the scale direction so that wrap-around at 360 degrees is handled:

```
v = v_{\min} + \frac{\theta - \theta_{\min}}{\theta_{\max} -
\theta_{\min}}\,\left(v_{\max} - v_{\min}\right)
```

For a non-linear scale, interpolate between the two nearest labelled major ticks. For a log scale, apply the same formula to log v. The scale information comes from the user, or from GenAI hook 1 after classical validation (see the hooks section).

Stage 8: Uncertainty and abstention

Four signals feed the confidence: the needle angle spread, the ellipse inlier ratio and fit residual, the tick comb residual, and how much glare overlaps the needle. The reading uncertainty is the angle spread passed through the scale slope, plus half of the smallest division:

```
u_v \approx \left|\frac{dv}{d\theta}\right|\,\sigma_\theta
```

For a linear scale the slope is (v_max - v_min) / (theta_max - theta_min). When any signal crosses its threshold, the system reports the reading as unreliable instead of guessing. Set the thresholds on the validation split, and evaluate the effect with a risk-coverage curve (error against the fraction of images answered).

Output: the value, its error bar and a status of reliable or unreliable.

Stages 9 to 11: optional extensions

The core project is Stages 1 to 8 plus the GenAI hooks; these three stages add recognition, repeatability and field use, and each can be dropped if time runs short.

Stage 9: Recognise the gauge type

The goal is to pick the right template (range, unit, scale type) when the photographed gauge belongs to a known set, and to find the gauge in a cluttered scene.

- **HOG with an SVM** detects the gauge region with a sliding window, the same method used for person detection.
- **LBP texture features** with an SVM classify the dial face by its printed texture.
- **Bag-of-Words** on SIFT descriptors with an SVM classifies the gauge type.

Compare the three on accuracy and a confusion matrix. When the user supplies the scale directly, this stage is not needed.

Stage 10: Register to a reference photo

For fixed installations, such as the same pump gauge photographed daily, store one reference photo with the dial region and scale marked. A new photo is matched to it with SIFT features, and RANSAC estimates the homography that warps the new photo into the reference frame. That gives a frontal view directly, so Stages 2 and 3 can be skipped and readings become more repeatable.

For a panel with several gauges, stitch overlapping photos into a panorama so that each gauge can be registered inside it. If the number of RANSAC inliers falls below a threshold, fall back to Stages 2 and 3.

Stage 11: Compression for upload

Sites with weak mobile networks need small images. Study how far the photo can be compressed before the reading degrades.

- Implement an 8x8 DCT, quantisation and run-length coding, and compare with a library JPEG encoder.

- Plot reading error against JPEG quality factor and compression ratio, and report PSNR against compression ratio.
- Report the lowest quality at which the reading stays within tolerance.

Where GenAI plugs in: three hooks

GenAI is used at three points, and each hook must beat a classical alternative in an ablation or be reported as not helping.

A hook is allowed only if it meets all three conditions:

1. It plugs into a step that is weak or manual in the classical pipeline.
2. Its output is validated by classical measurements before it is used.
3. An ablation measures its benefit, and the pipeline still runs with the hook switched off.

Hook	Where it plugs in	What it replaces	How it is tested
1. Scale parser	Before Stage 7	Manual entry of range, unit and scale type	End-to-end error with parsed vs hand-entered scale
2. Gross-error sentinel	After Stage 7	Nothing: the pipeline cannot tell when it is badly wrong	Gross errors caught vs false alarms, risk-coverage curve
3. Repair loop	After Stage 8, low-confidence images only	Fixed parameters and hand-written fallbacks	Images fixed vs a rule-based fallback and a grid search at equal compute

Hook 1: Scale parser

On MeasureBench, VLMs recognised units with over 90% accuracy but read values at about 30%, while an earlier OCR-based pipeline recognised units on only 17.8% of real images. A VLM is therefore a good reader of labels and a poor reader of needles.

- **Input:** the rectified dial face from Stage 3.
- **Fixed output fields:** unit; minimum and maximum value; scale type (linear, log, non-linear or dual); direction of increase (clockwise or anticlockwise); printed labels with approximate positions.
- **Classical validation:** match each label to the nearest detected major tick. Require the values to be monotonic along the scale direction, and for a linear scale require them to be evenly spaced (a straight-line fit with a small residual). On failure, reject the parse

and ask the user to enter the scale.
- **Test:** parse success rate, label-to-tick match rate, and end-to-end error with the parsed scale vs a hand-entered scale (the upper bound) vs no scale.

Hook 2: Gross-error sentinel

The VLM looks at the original photo and gives a coarse reading with its unit. If it differs from the geometric reading by more than a threshold (a fraction of full scale), the image is flagged. The VLM never replaces the classical value.

- **What it can catch:** tip and tail confusion, a flipped direction, the wrong scale ring, a wrong centre.
- **Threshold:** set it on the validation split. A starting value of 15% of full scale is a suggestion only.
- **Open question:** MeasureBench scores exact interval matches, so it does not say how often a VLM is roughly right. Whether a coarse VLM reading is good enough to flag gross errors has to be measured on your own data.
- **Test:** recall (the share of classical gross errors above 10% of full scale that get flagged), the false-alarm rate on correct readings, and whether the sentinel adds anything beyond Stage 8's own confidence, shown as risk-coverage curves with and without it.

Hook 3: Repair loop

The loop runs only when Stage 8 says the reading is unreliable or the sentinel flags the image.

- **What the VLM sees:** three overlays (the fitted ellipse on the photo, the unwrapped strip with detected ticks, and the needle mask) together with the numeric diagnostics.
- **What it may do:** pick one action from a fixed menu: stronger glare mask, different Canny thresholds, refit the ellipse on the inner ring, use the Hough line instead of the projection peak, swap tip and tail, or give up. The pipeline then reruns, for at most two rounds.
- **What it may not do:** it never outputs a reading.
- **Baselines:** no repair; a rule-based fallback that tries the same actions in a fixed order; a grid search over the same actions that picks by confidence score.
- **Success criterion:** it fixes more flagged images than the rule-based fallback at equal compute. If it does not, report that as a negative result.

What GenAI will not do

- Read the needle. VLMs are weakest at exactly this.
- Write reports or chat. That would not improve a single reading.

- Pass a number to the output without classical validation.

Cost and logging

Hook 1 runs once per gauge type, hook 2 once per image, and hook 3 only on flagged images. Any VLM API or an open-weight VLM works. Log every call (prompt, output and validation result) so you can report accuracy, calls per image and cost. Photos of real facilities can be sensitive, so use only your own images for demos that call an external API.

Dataset and ground truth

Collect about 280 photos of single-needle circular dials with known true readings, add synthetic degradations for graded difficulty, and test on the public MeasureBench dial subset.

Your own photos

- **Gauges:** about 10 single-needle circular types, for example a kitchen scale, sphygmomanometer, tyre pressure gauge, wall thermometer, hygrometer, barometer, oven thermometer, pressure-cooker gauge, analogue multimeter and a vehicle speedometer. Clocks and multi-needle meters are out of scope.
- **Base set:** about 10 readings per gauge at different needle positions in good conditions, which is about 100 photos.
- **Variation set:** for 3 gauges, 10 readings each at 3 tilt levels (0, 20 and 40 degrees) and 2 lighting conditions (room light and window glare), which is 180 photos.
- **Phones:** use at least 2 different phones.

Ground truth

- Kitchen scale: known weights.
- Thermometer: a digital reference thermometer in the same water.
- Other gauges: two people read independently to the smallest division, and a third reading settles any disagreement.
- For every photo, record the true value and its uncertainty, which is half of the smallest division.

Synthetic degradation

Take the clean photos and add Gaussian blur at several strengths, linear motion blur, Gaussian and salt-and-pepper noise, several JPEG quality levels, and brightness and contrast shifts. This gives graded difficulty with the same ground truth and no extra

photography.

Splits

- **Validation split:** used only to set thresholds (the abstention thresholds and the sentinel threshold).
- **Test split:** touched once, at the end.
- Hold out at least two gauge types entirely for testing, to measure generalisation to unseen gauges.

Public data: MeasureBench

- MeasureBench on Hugging Face has 1,272 real-world images and 1,170 synthetic test images under the CC BY-SA 4.0 licence. Its real subset has 711 dial images, which include clocks and multi-needle instruments, so filter by hand to single-needle circular dials.
- It is scored with valid reading intervals, so use those intervals as ground truth and also report numeric error.
- The dataset repository also lists Blender files that can be used to generate synthetic gauges.
- Optional: the 4,813-image real set from Leon-Alcazar et al. Check its availability and licence before relying on it.

Experiments and metrics

Six experiments measure every stage and every GenAI hook, with error reported as a percentage of full scale (FS).

Metrics

```
e = \frac{\left|v_{\text{pred}} - v_{\text{true}}\right|}{v_{\max} - v_{\min}} \times 100\%
```

- **Primary:** the error e above. Report the mean, median and 90th percentile.
- **Gross-error rate:** the share of images with e above 10% of FS.
- **Within-tolerance rate:** the share of readings within one smallest division, which is comparable to MeasureBench's interval matching.
- **Coverage and risk:** coverage is the share of images answered, and risk is the error rate on the answered ones. Plot risk against coverage.
- **Cost:** VLM calls per image and run time per image.

- With about 300 images, report bootstrap confidence intervals.

Experiments

ID	Question	Method
E1	Does each classical stage matter?	Switch off one stage at a time (restore, tilt correction, polar unwrap, DFT tick check, uncertainty) and tabulate the error
E2	How does error change with tilt, glare, blur, noise and JPEG quality?	Vary one factor at a time using the variation set and the synthetic degradations, and compare with a VLM alone
E3	Does each GenAI hook help?	The five-step ablation ladder below
E4	Does the sentinel add anything beyond Stage 8′s own confidence?	Risk-coverage curves with and without the sentinel
E5	Does the pipeline generalise to unseen gauges?	Error per held-out gauge type
E6	How does it compare with VLMs on public data?	Run on the single-needle dial subset of MeasureBench and compare with its published numbers, noting the different scoring and the older models

Optional: report the Stage 9 classifier accuracy with a confusion matrix, and the Stage 11 compression curves.

GenAI ablation ladder (E3)

Configuration	What is switched on	Mean error (% FS)	90th percentile	Gross-error rate	Coverage at fixed risk	VLM calls per image
C0	Classical pipeline, scale entered by hand	TBD	TBD	TBD	TBD	0
C1	C0 plus hook 1 (scale parser)	TBD	TBD	TBD	TBD	TBD
C2	C1 plus hook 2 (sentinel flags)	TBD	TBD	TBD	TBD	TBD

Configuration	What is switched on	Mean error (% FS)	90th percentile	Gross-error rate	Coverage at fixed risk	VLM calls per image
C3	C2 plus hook 3 (repair loop)	TBD	TBD	TBD	TBD	TBD
B	VLM alone, as the baseline	TBD	TBD	TBD	TBD	1

Success criteria

Write the criteria down before running the test split. A hook counts as helpful if it lowers the gross-error rate at equal coverage, or raises coverage at equal risk, by more than the bootstrap confidence interval. As a suggested starting point, hook 2 helps if it flags at least half of the gross errors with a false-alarm rate under 10%; tune these numbers on the validation split. A hook that misses its criterion is reported as not helping.

Week-by-week plan

Each of the 14 lab programs doubles as one module of DialReader, so every week's submission is also a step in the project.

Week	Lab program	DialReader module	Output
1	SciPy and matrix arithmetic	Matrix toolkit: rotation, affine and homography matrices, polar sampling grid; repository and data loader skeleton	Utility library with tests
2	Image attributes and negation	Dataset loader, image attributes (size, channels, histogram, EXIF), negation for dark-on-light and light-on-dark needles	Dataset statistics table
3	2D and 3D transformations	Tilt correction (affine and homography) and polar unwrap, Stages 3 and 4	Rectified and unwrapped strips
4	Enhancement with filters and histograms	Stage 1 contrast, smoothing and sharpening	Before and after images, histograms

Week	Lab program	DialReader module	Output
5	Linear and non-linear filters	Stage 1 noise and blur: Gaussian, median, Wiener, plus glare handling	Noise and blur robustness table
6	Image segmentation	Needle mask and glare mask with Otsu and region-based segmentation	Needle masks
7	Image compression	Stage 11: DCT, quantisation and run-length coding, error against quality	Compression curves
8	Morphological opening and closing	Mask cleanup and needle thinning	Clean needle skeleton
9	Canny edge detection	Canny, Hough circle and line, RANSAC ellipse, tick detection with DFT; first end-to-end version	Classical v1 reads clean photos (Stages 1 to 8)
10	HOG for person detection	Stage 9: HOG and SVM gauge detector	Detection precision and recall
11	SIFT interest points	Stage 10: SIFT features for registration to a reference photo	Feature matches
12	Image stitching panorama	RANSAC homography registration and panel stitching	Registered or stitched gauge panel
13	LBP	Stage 9: LBP gauge-type classifier	Classifier accuracy
14	Image classification with Bag of Words	Bag-of-Words gauge-type classifier, final integration and ablations	Confusion matrix and final results

Start collecting photos in weeks 1 to 4 so the data is ready when Stage 9 and the experiments need it. The GenAI hooks are not a lab program: build them alongside weeks 10 to 14, after the classical version works at week 9.

Milestones

1. **After week 5:** Stage 1 works, with robustness numbers on synthetic degradations.

2. **After week 9:** the classical pipeline reads clean photos end to end, with an error bar.
3. **After week 12:** registration and gauge-type recognition work, and hook 1 is integrated.
4. **Week 14:** all three hooks, experiments E1 to E6, the demo and the report are done.

Deliverables

- A code repository with one module per week.
- A Streamlit demo: upload a photo, see every stage, and get the reading with its error bar and a reliable or unreliable flag.
- The dataset with its collection protocol and ground truth.
- A report with experiments E1 to E6, including the GenAI ablation table and the negative results.

Syllabus coverage

DialReader uses topics from all five units, and a few topics it does not use are listed under the table.

Unit	Syllabus topic	Where it is used
1	Sampling, quantisation, pixel relationships	Polar resampling with interpolation, JPEG quantisation, connectivity in mask cleanup
1	Imaging geometry, 2D and 3D transforms	Ellipse-to-circle tilt correction, homography, polar transform
1	Colour imaging	Colour-channel separation of a red needle
1	DFT	Tick periodicity in Stage 5, blur-kernel check in Stage 1
1	DCT	Compression study in Stage 11
1	Haar transform	Optional denoising of the tick profile
2	Histogram processing	Stage 1 contrast
2	Spatial smoothing and sharpening filters	Stage 1
2	Frequency-domain filtering	Spectrum analysis of the tick profile; optional notch filter for periodic print patterns
2	Dilation, erosion, opening, closing	Glare mask, needle cleanup, thinning

Unit	Syllabus topic	Where it is used
3	Restoration: degradation model, inverse and least-mean-square filters	Stage 1 deblurring with a Wiener filter
3	Compression: DCT, run-length coding, JPEG	Stage 11
3	Segmentation: thresholding, region-based	Needle and glare masks
4	Edge detection: gradient, Canny, DoG	Stage 2; DoG optional for tick scale
4	LoG blob detection and Harris corners	Optional independent check of tick ends
4	Hough transform	Dial circle and needle line
5	SIFT and interest points	Registration in Stage 10
5	RANSAC	Ellipse fit in Stage 2, homography in Stage 10
5	HOG	Gauge detector in Stage 9
5	LBP	Gauge-type classifier
5	Bag-of-Words	Gauge-type classifier
5	Image stitching	Panels with several gauges

Not used: the hit-or-miss transform and arithmetic and LZW coding. Huffman coding can be added to the Stage 11 compressor if you want it.

Risks and what is not yet verified

The biggest risks are a classical pipeline that fails on unseen gauges and a GenAI hook that does not help, and the plan handles both by measuring and reporting honestly.

Risk	Why it matters	Mitigation
Classical pipeline fails on unseen gauge types	Earlier special-purpose systems scored very low on MeasureBench's unseen gauges	Restrict scope to single-needle circular dials, test on held-out gauge types, abstain when unreliable, report failures openly

Risk	Why it matters	Mitigation
Glare and reflections on the cover glass	Can hide the needle or break the ellipse fit	Glare mask, lower confidence when glare overlaps the needle, retake-the-photo flag
Non-standard dials (semi-circular, vertical, dual scale)	The calibration formulas assume a circular arc	Out of scope at first; hook 1 reports the scale type so unsupported types are rejected
Multi-needle and composite instruments	Clocks and water meters scored near zero for every model on MeasureBench	Out of scope
A hook does not help	A decorative component weakens the project	Write success criteria first, keep the pipeline working with hooks off, report negative results
Weak ground truth	Wrong labels hide real errors	Two independent readers, recorded uncertainty, reference instruments where possible
API cost, availability and privacy	Repeated VLM calls cost money and photos can be sensitive	One call per gauge type for hook 1, call hook 3 only when flagged, cache results, use only your own images for external APIs, consider an open-weight VLM
Overclaiming novelty	The prior-work search was not exhaustive	Say "no prior work found" and do a proper literature check before submission
Use in safety-critical settings	A wrong reading could mislead	Always show the error bar and the reliability flag, and state that the tool is not for safety-critical decisions
Scope creep	Stages 9 to 11 take time	Treat them as optional; the core is Stages 1 to 8 plus the hooks

Not yet verified

- Prior work on the three specific GenAI hooks was not searched; only gauge reading in general was.
- The 2026 gauge-video study was read from its abstract only.
- The MeasureBench results are for 2025 models, so newer VLMs may score higher.
- The availability and licence of the Leon-Alcazar real dataset were not checked.
- Whether a VLM's coarse reading is accurate enough for the sentinel (hook 2) is unknown until measured on your data.

Sources

Sources were checked on 12 October 2026. Only the MeasureBench paper was read in full; the others were read from search-result excerpts and abstracts.

- MeasureBench: Do Vision-Language Models Measure Up?
- MeasureBench dataset on Hugging Face
- Lost in Motion: Vision Language Models Fail the Dynamic Gauges Test
- Reitsma et al., Under pressure: learning-based analog gauge reading in the wild
- Leon-Alcazar et al., Learning to Read Analog Gauges from Synthetic Data
- Benchmarking Vision-based Analog Gauge Reading Methods for Vehicle Dashboard Validation
- DialBench: pointer meter reading with large foundation models
- Type-Balanced Federated Learning for Visual Analog Meter Reading
- Automated river gauge plate reading with object detection and generative AI